

## NORCET Ki POL



### Contraceptive Methods

#### Ideal Contraceptive

An ideal contraceptive should have the following characteristics:

- ✓ **Safe** - Minimal or no side effects
- ✓ **Effective** - High efficacy in preventing pregnancy
- ✓ **Affordable** - Low cost / economical
- ✓ **Easily available** - Accessible to the population
- ✓ **Easy to use** - Simple method, minimal training required
- ✓ **Acceptable** - User compliance and motivation is high
- ✓ **Easy to maintain** - Minimal follow-up and maintenance
- ✓ **Reversible** - Rapid return of fertility after discontinuation

#### ✂ **NORCET Key Point:**

No single method fulfills all the criteria → *Ideal contraceptive does NOT exist.*

#### Eligible vs Target Couple

##### **Eligible Couple**

- Female age 15-49 years
- IDEAL METHOD: Temporary contraception

##### **Target Couple**

- Completed family
- ≥1 living child
- IDEAL METHOD: Permanent sterilization

## CLASSIFICATION OF CONTRACEPTION

### TEMPORARY / REVERSIBLE

- Natural methods
- Barrier methods
- IUD
- OCP
- Injectables
- Implants
- Emergency contraception

### PERMANENT

- Male: Vasectomy
- Female: Tubectomy

### NATURAL METHODS

#### Lactational Amenorrhea

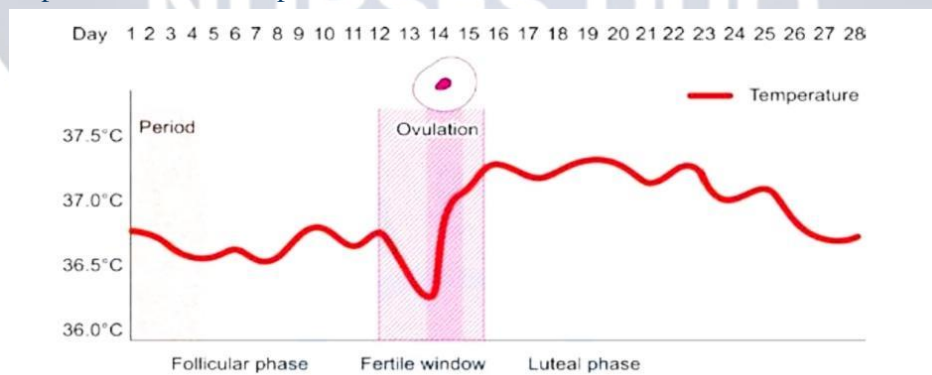
- *Bellagio Criteria: EBF for 6months > Amenorrhea*
- Effective only in early postpartum
- High failure rate
- *Best during breastfeeding → Progesterone-only pill (POP)*

#### Rhythm Method

- Based on menstrual calendar

#### Basal Body Temperature

- Rise of 0.3-0.5°C after ovulation
- Unsafe period → After temperature rise

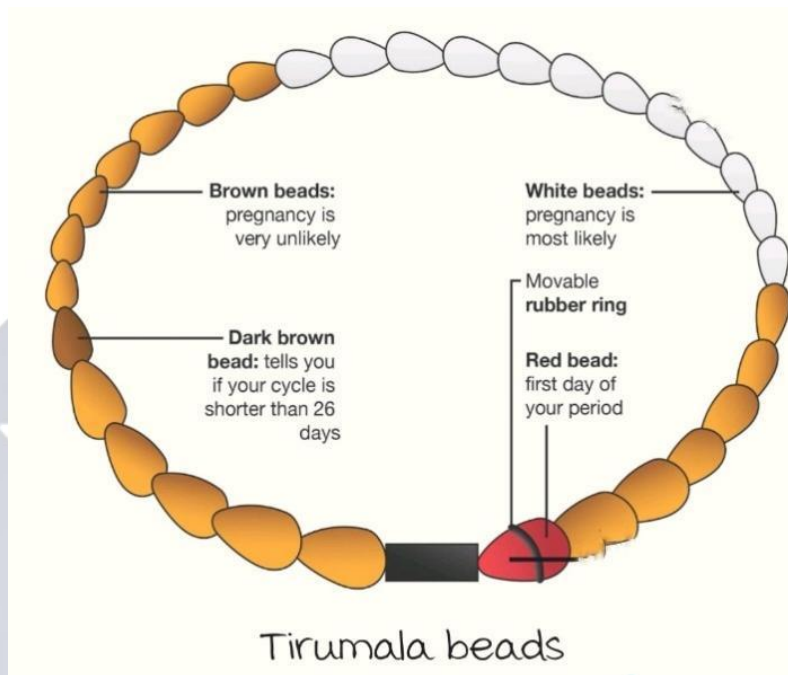


#### Cervical Mucus (Billings)

- Thin, clear mucus → fertile period
- Very wet, stretchy, resembles raw egg white: ovulation

#### Standard Days Method ( Tirumala Beads)

- For regular cycles only
- Not used if cycle <26 or >32 days
- Useful in illiterate women



Red bead: First day of menses  
White beads: Unsafe period

### BARRIER METHODS

#### **Male Condom**

- Latex
- Prevents pregnancy + STDs



#### **Female Condom**

- Polyurethane or latex



### **Cervical diaphragm**

- A cervical diaphragm is a flexible, dome-shaped barrier device made of latex or silicone that is placed inside the vagina to cover the cervix, preventing sperm from entering the uterus.
- It is usually used with a spermicide.
- Reusable and cost-effective
- Can be inserted before sexual intercourse
- Must be left in place for at least 6 hours after intercourse
- Risk of toxic shock syndrome if left too long



### **Cervical Cap**

Small, cup-shaped barrier device made of rubber or silicone that fits snugly over the cervix, blocking sperm entry into the uterus.



## INTRAUTERINE DEVICES (IUDS)

### Generations

- 1st Gen (Inert)  
Lippes loop

### 2nd Gen (Copper)

- CuT 200, CuT 380A, Cu 375

### 3rd Gen (Hormonal)

- Levonorgestrel-releasing iud, Mirena, Progestasert

## HORMONAL IUDs

| Progestasert       |                                                                                                                                 | Mirena                                                                                                                            |  |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--|
| Contains           | Natural progestin                                                                                                               | Levonorgestrel/LNG-20 (Synthetic)                                                                                                 |  |
| Life Span          | 1 year                                                                                                                          | 5 years                                                                                                                           |  |
| Total hormone load | 38 mg                                                                                                                           | 52 mg                                                                                                                             |  |
| Rate of secretion  | 65 mcg/day                                                                                                                      | 20 mcg/day                                                                                                                        |  |
| Characteristics    | <ul style="list-style-type: none"> <li>✓ Lowest expulsion rate</li> <li>✓ ↑ ectopic pregnancy if contraception fails</li> </ul> | <ul style="list-style-type: none"> <li>✓ Lowest failure rate (&lt;0.3 / 100 women years)</li> <li>✓ Lowest side effect</li> </ul> |  |

## COPPER IUDS

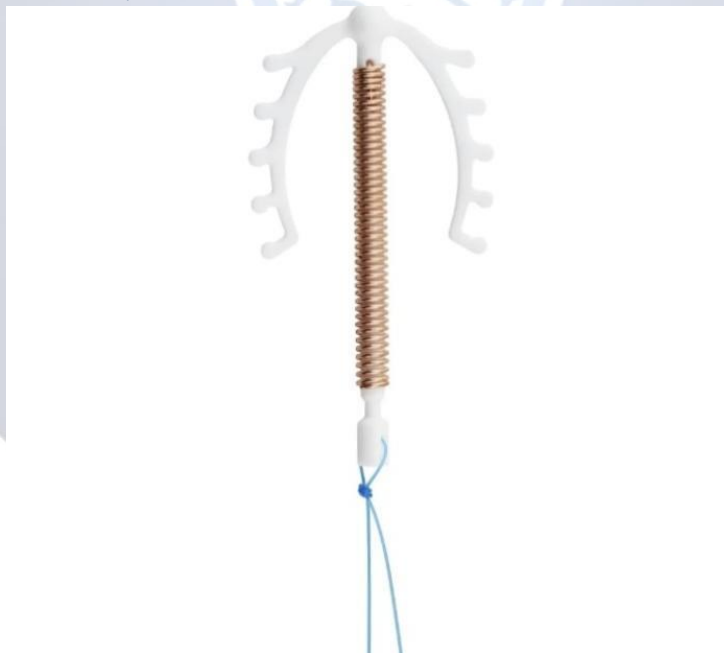
### **CuT 380A**

- T-shaped
- Duration: 10 years
- Copper on arms + stem



### **Cu 375**

- U-shaped
- Duration: 5 years
- Copper on stem only



| • CuT380A                                                                                                                                                                                                                                                                                                                                                          | • Multiload                                                                                                                                                                                                                                                                                                                                                    | • CuT200                                                                                                                                                                                                                                      | • Copper 7                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>T-shaped frame (Ba sulfate for X-ray opacity)</li> <li>314 mm<sup>2</sup> + 33 mm<sup>2</sup> on each arm = 380 mm<sup>2</sup></li> <li>Insertion withdrawal technique</li> <li>Supplied free of cost by GOI</li> <li>10 years</li> </ul>  | <ul style="list-style-type: none"> <li>Copper on the vertical shaft only</li> <li>The 2 arm are flexible and serrated</li> <li>No plunger</li> <li>Pre-loaded withdrawal inserters</li> <li>250 mm<sup>2</sup> / 375 mm<sup>2</sup></li> <li>Cu 375: GOI, 5 years</li> </ul>  | <ul style="list-style-type: none"> <li>Gyne-T</li> <li>No longer supplied by GOI</li> <li>200 mm<sup>2</sup> copper</li> <li>Life span: 3 years</li> </ul>  | <ul style="list-style-type: none"> <li>Gravigard</li> <li>Shape of number 7</li> <li>200 mm<sup>2</sup></li> <li>3 years</li> </ul>  |

### Mechanism

- Copper → spermicidal
- Prevents fertilization > implantation

### IUD Insertion Timing

- Post-placental: within 10 min of delivery
- Postpartum: 10 mins to 48 hours (upto 6 weeks)
- Interval: after 6 weeks
- Post-abortion: after 12-15 days
  - 12 days: Surgical abortion
  - 13-15 days: Medical abortion

### IUD Contraindications (Absolute)

- Pregnancy
- Active pelvic infection
- Undiagnosed vaginal bleeding
- Genital malignancy

⚠ Previous ectopic pregnancy = NOT a contraindication

### IUD Side Effects

- Most common → Bleeding/spotting
- Pain
- PID

- Perforation (rare)

#### NORCET CASE BASED QUESTION:

A woman comes to OPD after 2 days of insertion of PPIUCD with complaints of bleeding.

What would you advise to the woman?

*Reassurance + Iron Folic Acid tablets*

### ORAL CONTRACEPTIVE PILLS (OCP)

**Mala-D, Mala-N:** Ethinylestradiol 0.03mg + Levonorgestrel 0.15mg

- 21 hormonal pills + 7 iron pills included
- Mala N: Free of cost by GOI
- Mala D: Home delivery by Asha worker



#### Non-Hormonal Oral Pill

##### **Centchroman (Ormeloxifene)**

- Sold under brand name Saheli/ Chhaya
- MOA: Selective estrogen receptor modulator

- *Non hormonal, non steroidal pills*
- Biweekly for first 3 months then weekly
- 30mg pills

**Side effects:**

- Menstrual irregularities
- Hot flushes
- Mood changes



MISSING A PILL (High Yield)

# NURSES DUO

BY NURSES, FOR NURSES 

**NORCET**

## MISSING A PILL: IMPORTANT



### One Missed Pill



- ✓ The missed pill should be taken as soon as possible
- ✓ The remaining pills are taken as usual  
No additional contraception required

### 2 or more missed pills (> 48 h late)



- ✓ The last missed pill should be taken as soon as possible
- ✓ Leave the earlier missed pills
- ✓ Use additional contraception for 7 days  
Further to reduce risk of pregnancy

- ✓ If pills missed in 1<sup>st</sup> week:
  - Consider Emergency Contraception
- ✓ Pills missed in 2<sup>nd</sup> week: No need for Emergency
  - Pills missed in the 3<sup>rd</sup> week:
    - Omit the pill free interval and start the next pack after finishing the active pills in the current pack

Follow Nurses\_duo\_ for more



### Emergency Contraception

- Ezy pill/ i pill/ Lng pill
- Levonorgestrel 1.5 mg
- Single dose within 72 hours of intercourse



### INJECTABLE CONTRACEPTIVES

#### DMPA: Depot Medroxyprogesterone Acetate

- Brand name: Antara
- Dose: 150 mg IM every 3 months
- Benefits: ↓ menstrual blood loss
- No additional contraceptive required
- Follow-up: 2 weeks before to 4 weeks after due date



#### NET-EN

- Norethisterone Enanthate
- Dose: 200 mg IM monthly

### VAGINAL RING

Small, flexible hormonal device inserted deep into the vagina that releases estrogen and progestin to prevent pregnancy.

**Composition:** Ethinyl estradiol, Etonogestrel

**Method of Use:**

- Kept in place for 3 weeks
- Removed for 1 week (withdrawal bleeding)
- New ring inserted after 7 days

**Mechanism of Action:**

- Suppresses ovulation
- Thickens cervical mucus
- Makes endometrium unfavorable for implantation

**Advantages:**

- Convenient and user-friendly
- Highly effective (failure rate <1% with ideal use)
- Regular and predictable menstrual cycles
- Reduced PMS symptoms
- Fewer systemic side effects than oral pills



### **TRANSDERMAL CONTRACEPTIVE PATCH**

Hormonal patch applied to the skin that releases estrogen and progestin

**Usage:**

- Applied once weekly for 3 weeks
- Followed by 1 patch-free week



### PHEXXI

A non-hormonal vaginal contraceptive gel used before intercourse.



### PERMANENT METHODS

#### Vasectomy

- Sterility achieved after: 3 months OR 30 ejaculations

- Cost effective

### Tubectomy

- Most common contraceptive method used in India (approx 36%)

### FAILURE RATE OF CONTRACEPTION

Best assessment method → Life-table analysis

Pearl Index formula: Total no. of accidental pregnancies × 100 / Women years of exposure

Pearl Index (EXAM FAVORITE)

- Condoms: 2-20
- IUD: 0.5-1.5
- OCP: <1 (best oral method)
- Implants: <0.05 (best overall)

### CONTRACEPTIVE OF CHOICE (COC) - VERY IMPORTANT

- ✓ **Breastfeeding women**: POP
- ✓ **Unmarried women**: Barrier + OCP
- ✓ **Spacing**: IUD / OCP
- ✓ **Occasional intercourse**: IUD (↓accidental pregnancy)

### Medical Eligibility Criteria (WHO)

- ✓ Category 1: Safe
- ✓ Category 2: Use with caution
- ✓ Category 3: Avoid unless necessary
- ✓ Category 4: Contraindicated

### NORCET ONE-LINERS

- ✓ Best reversible method → Implants
- ✓ Best during lactation → POP
- ✓ Most common IUD side effect → Bleeding
- ✓ IUD prevents mainly → Fertilization
- ✓ Sterility after vasectomy → 3 months
- ✓ Copper number indicates → Surface area of copper

### MISCELLANEOUS

### Mission Parivar Vikas (MPV) -NORCET Focus

**Aim:** Improve family planning services in\_high-fertility districts by promoting modern contraceptive methods.

#### **Contraceptives promoted under MPV**

- Injectable contraceptive: Antara (DMPA)
- Weekly oral pill: Chhaya (Centchroman)
- Daily oral pill: Mala-N
- Progesterone-only pill: Ezy pill
- Barrier method: Nirodh (condom)
- Intrauterine devices: IUCD / PPIUCD
- Permanent methods:  
    *Female sterilization*  
    *Male sterilization*

### Nishchay - High Yield

*Provides free pregnancy testing kits at community level (PYQ)*

Promotes Fixed Day Sterilization Services

#### Fixed Day Approach

- PHC: Monthly
- CHC: Fortnightly
- Sub-district hospital: Weekly
- District hospital: Twice weekly

*Exam Pearl: Nishchay = early pregnancy detection + assured sterilization services.*

